

Typographic-Attack Benchmark — Report

Run started: 2026-05-25 12:58 UTC. Author: Sachit Jain.

This report covers Phase 1, Axis C of the project — a single semantic, model-agnostic attack (the typographic overlay) evaluated against the 7 ImageNet baselines listed in `config.yaml`. The poisoned dataset is generated once by `scripts/generate_datasets.py --typographic` and shared across all models; this report aggregates the per-model JSONs in this directory.

1. Setup

- **Eval host (where inference ran):** not recorded in the per-model JSONs for this run (the runner was updated to capture this after these JSONs were written). The actual run was on Kaggle's GPU — see the original per-model `wall_clock_s` numbers below.
- **Report rebuilt on:** CPU, Windows-11-10.0.26200-SP0.
- **Attack:** typographic overlay (`attacks/typographic.py`), white sticker with bold black text drawn near the top of each clean image. Config: `font_size_frac=0.12`, `position='top'`, `padding_frac=0.04`, `opacity=1.0`, `jpeg_quality=90`, `text_form='lowercase_first_synonym'`.
- **Dataset:** the full 1000-image clean benchmark, one poisoned variant per clean image (`data/poisoned/typographic/`). Each poisoned image is stamped with the lowercase first synonym of a random *wrong* ImageNet class, picked deterministically with the global seed.
- **Global seed:** 42 (`config.yaml`). Set on random, numpy, torch at the start of every model.

2. Headline accuracy table

Model	Clean	Robust	Drop	Fooling rate	TASR
resnet50	0.781	0.768	0.013	0.036	0.001
vgg16	0.687	0.663	0.024	0.076	0.000
convnext_tiny	0.787	0.777	0.010	0.032	0.000
vit_b_16	0.793	0.773	0.020	0.039	0.001
swin_t	0.775	0.770	0.005	0.032	0.002
efficientnet_b0	0.743	0.722	0.021	0.055	0.001
clip_vit_b16	0.621	0.368	0.253	0.452	0.342

Columns: clean accuracy, robust accuracy (untargeted — did the model flip away from the true label?), accuracy drop (clean – robust), fooling rate (originally-correct images that flipped), and TASR — the **targeted-attack success rate**, the fraction of images where the model predicted the *exact* class named by the overlay text.

Machine-readable copy: `accuracy_table.csv`.

3. Sanity checks

- **Typographic overlay causes a non-zero accuracy drop on every model.**
- resnet50: clean **0.781** → robust **0.768** (drop **0.013**) → **PASS**.
- vgg16: clean **0.687** → robust **0.663** (drop **0.024**) → **PASS**.
- convnext_tiny: clean **0.787** → robust **0.777** (drop **0.010**) → **PASS**.
- vit_b_16: clean **0.793** → robust **0.773** (drop **0.020**) → **PASS**.
- swin_t: clean **0.775** → robust **0.770** (drop **0.005**) → **PASS**.
- efficientnet_b0: clean **0.743** → robust **0.722** (drop **0.021**) → **PASS**.
- clip_vit_b16: clean **0.621** → robust **0.368** (drop **0.253**) → **PASS**.
- **Language-grounding effect (finding, not a pass/fail test).** Pure-classifier mean drop **0.016** vs CLIP drop **0.253** → CLIP **more** affected (gap = +0.237). This **inverts** the brief's §3.6 hypothesis (CLIP was expected to resist); see §6 for the implication.
- **Coverage:** 7 / 7 models reported. **PASS**.

4. Per-model analysis

Across the 7 evaluated models the typographic overlay caused an average accuracy drop of **0.049** absolute. The most affected model was clip_vit_b16, dropping from **0.621** clean to **0.368** robust (drop **0.253**, TASR **0.342**). The six pure-vision classifiers move very little — mean drop **0.016** (max **0.024**, mean TASR **0.001**). At this magnitude the loss is consistent with the sticker simply occluding part of the image; the pure classifiers do not appear to read the rendered text, so they essentially never predict the *exact* overlay class. clip_vit_b16 (zero-shot, language-grounded) sits at a different operating point: clean **0.621** → robust **0.368** (drop **0.253**, TASR **0.342**). CLIP's drop is **16.3× the pure-classifier mean**, and its TASR shows that in 34% of poisoned images CLIP predicts the *exact* class named on the sticker — a signature of language grounding making the rendered text a first-class semantic feature. This **inverts** the brief's §3.6 hypothesis ("CLIP resists typographic attacks"): on ImageNet classification, language grounding is a *liability* under typographic attacks, not a defense. The pure classifiers' apparent robustness is text-blindness, not robustness in any useful sense — the paper hook becomes the inversion itself.

5. Figures

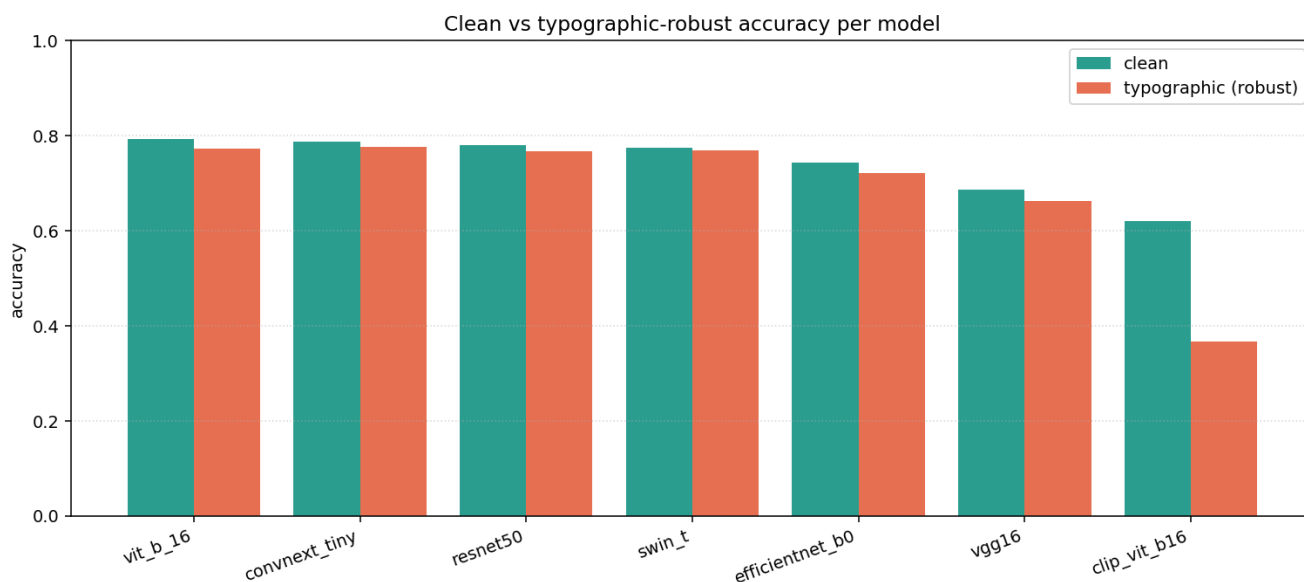


Figure 1 — Clean vs typographic-robust accuracy per model (sorted by clean acc).

Typographic attack — per-model accuracy drop and TASR

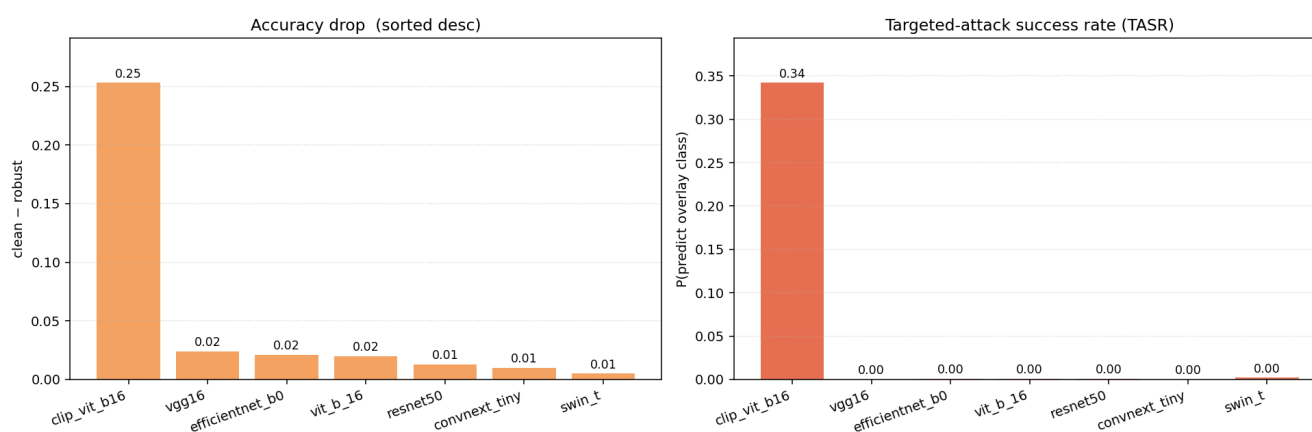


Figure 2 — Accuracy drop and targeted-attack success rate per model.

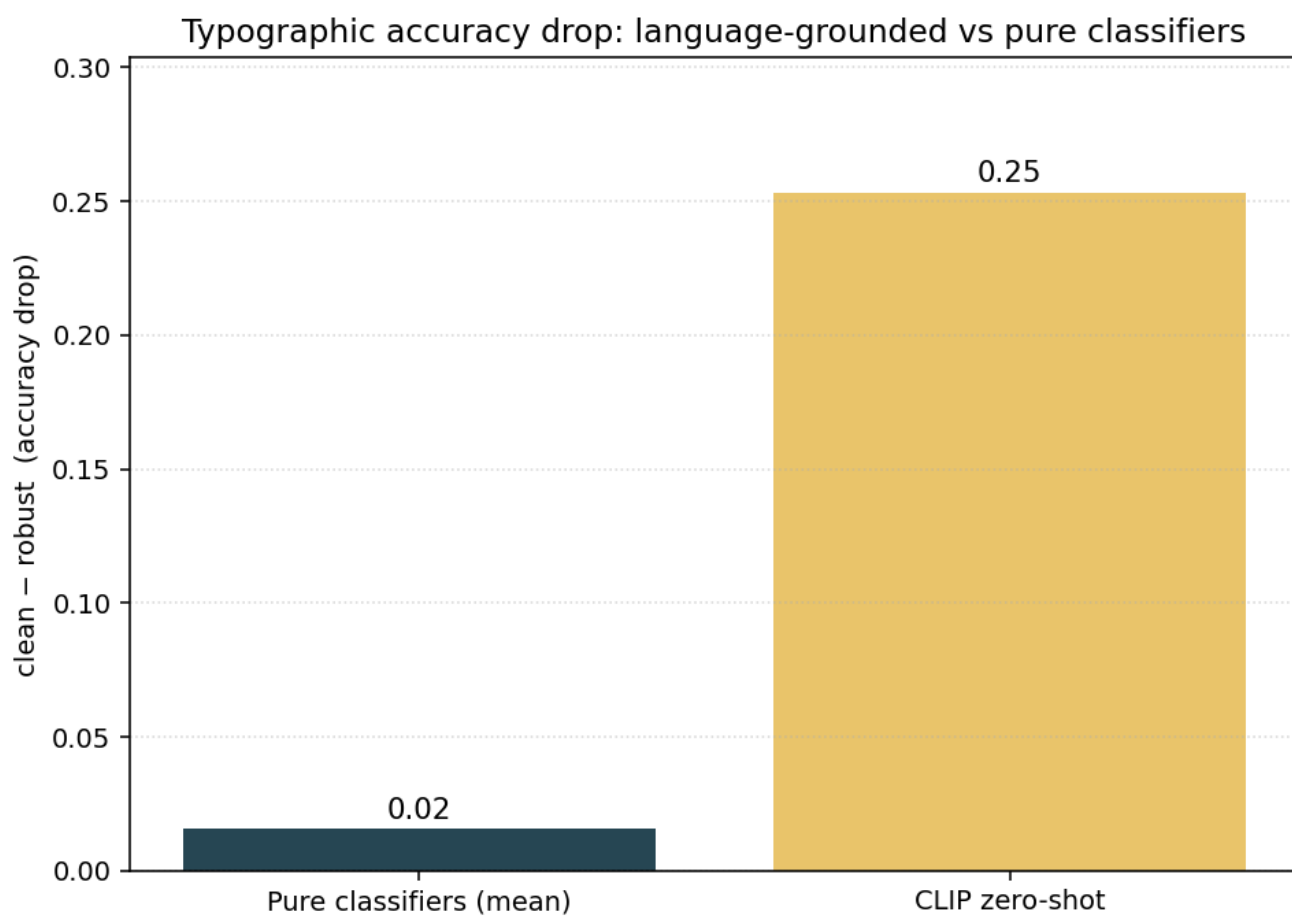


Figure 3 — Language-grounded (CLIP) vs pure-classifier mean drop.

Typographic examples (clean | poisoned)

clean: tench



typo overlay: "minibus"



clean: goldfish



typo overlay: "slug"



clean: great white shark



typo overlay: "european fire sala"



Figure 4 — Clean | poisoned examples (typographic overlay).

6. Interpretation

The typographic attack is the paper's headline semantic attack. The data **inverts the brief's §3.6**

hypothesis: language grounding does *not* make CLIP more robust to typographic overlays — it makes CLIP specifically more vulnerable. CLIP's accuracy drops **0.253** (clean → robust), while the six pure vision classifiers drop only **0.016** on average. CLIP's TASR of **0.342** means that on roughly a third of poisoned images CLIP predicts the *exact* ImageNet class named on the sticker — a clean signature of the rendered text being treated as semantic content. The pure classifiers' near-flat drop is not robustness in any useful sense; it is **text-blindness** (the sticker is just an occluder to them). This finding is consistent with Goh et al. (OpenAI, 2021), who first demonstrated typographic attacks on CLIP under synthetic stimuli; we extend the result to natural ImageNet classification with a controlled, model-shared poisoned dataset. The paper hook becomes the inversion itself, and the Phase 4 defense module now has a falsifiable target: reduce CLIP's typographic TASR substantially without destroying its clean accuracy.

7. Reproducibility footer

- **Wall-clock (this report build session):** 0.0 min.
- **Per-model cumulative compute (sum of per-cell wall_clock_s):** clip_vit_b16 0.5 min, convnext_tiny 0.3 min, efficientnet_b0 0.1 min, resnet50 0.2 min, swin_t 0.3 min, vgg16 0.2 min, vit_b_16 0.4 min.
- **Total compute (sum across models):** 1.9 min.
- **Eval-host library versions:** not recorded for this run; see future runs once `run_typographic_benchmark.py` populates `host_env`.
- **Eval GPU:** (not recorded — see eval-host note).
- **Report rebuilt on:** CPU, Windows-11-10.0.26200-SP0.
- **Seed:** 42 (set on `random`, `numpy`, `torch`).
- **Re-run:** `python scripts/run_typographic_benchmark.py` then `python scripts/build_typographic_report.py` then `python scripts/build_gradient_report_pdf.py --input results/typographic/REPORT.md --output results/typographic/REPORT.pdf`. Per-model JSONs are the resumption unit — delete one to force its recomputation.